

Yongxue Xu (Jerry)

Undergraduate researcher at Sun Yat-sen University

Video generation • World models • 4D scene understanding

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Research interests span video understanding and generation, multimodal foundation models, and world models, with a focus on spatiotemporal modeling, 4D dynamic reasoning, long-horizon prediction, and physical consistency. Further interests include unified understanding and generation, and perception and interaction for embodied intelligence.

Education

Sun Yat-sen University Project 985 / 211 / Double First-Class
B.Eng. in Intelligent Science and Technology

SEP 2023 -- JUN 2027
GPA 3.91 / 4.0 Major GPA 3.94 / 4.0

Research Experience & Publications

LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue

Chaoyue Li*, **Yongxue Xu***, Jie Feng*, and Jiayu Ding†

Submitted to NeurIPS 2026; [arXiv:2605.19390](https://arxiv.org/abs/2605.19390)

Developed [LMM-Track4D](#) for trajectory-grounded multi-turn dialogue: answering spatiotemporal questions about multi-view videos while predicting query-conditioned 3D object trajectories. Combined geometry-aware encoding, persistent state memory, and structured trajectory decoding; co-developed Track4D-Bench.

[Equipped LMMs with 4D dynamic reasoning capabilities. Substantially outperformed the base model on Track4D-Bench and external benchmarks, especially in dynamics reasoning.](#)

Contribution: Led model design and main experiments; co-developed the benchmark.

Learning to Reason with Persistent Object States for Video Instance Segmentation

Yongxue Xu*, Boxue Yang*, Ziqian Liu, Shaoqiu Zhang, Rui Qian, and Linfeng Zhang†

Submitted to ICLR 2027

Developed [POSReasoner](#) for video instance and object segmentation, preserving identities as objects become occluded and reappear. Coupled persistent object memory with verified state transitions to prevent ambiguous observations from corrupting identity history.

[Improved long-horizon identity continuity and object recovery with the segmentation models frozen. Delivered consistent gains across evaluated backbones and benchmarks, including occlusion-heavy and long-video settings.](#)

Contribution: Led method and architecture design, implementation, experiments, and manuscript preparation.

TRACE: Typed Reference-Aware Conversational Editing for Humanoid Motion Diffusion

Zejian Kang, Yifu Luo, Boyang Shen, Jingxuan Li, Zekai Chen, Shi Zekun, **Yongxue Xu**, Zhenyu Hou, Bingqian Wu, Shaolei Zhang, Wentao Yang, and Xiangru Huang

Submitted to ICLR 2027

Contributed to TRACE for multi-turn, language-guided humanoid motion generation and editing. Typed motion programs specify what to change or preserve, while reference-constrained diffusion modifies selected joints and time intervals without regenerating the entire motion.

Contribution: Reproduced baseline methods, adapted evaluators, and converted motion demos for Unitree G1.

LaST: Latent State Tracking for Persistent Object Memory in Video World Models

Yongxue Xu and Harry Yang[†]

Under review at ICLR 2027

LaST addresses object reappearance after occlusion or leaving the view in long-video generation. Fixed-size scene and object memory preserves geometry and appearance beyond the local context, guiding a frozen generator for consistent reappearance without caching the full history.

Improved all five WRBench metrics over Short-KV and Full-KV at both 1.3B and 5B scales.

Graph Domain Adaptation Does Not End with Representation Learning

Ziqian Liu^{*}, Yongxue Xu^{*}, Enze Zhang, Jiaqi Zhang, Hao Wang, and Maolin Wang[†]

Submitted to ICLR 2027; [arXiv:2609.25692](#)

Developed **EviGDA** for unsupervised graph domain adaptation: transferring node classifiers from labeled source graphs to unlabeled target graphs by fusing graph-aware and feature-only predictions.

Surpassed existing state-of-the-art methods in overall graph domain adaptation performance.

Contribution: Designed the core method and led implementation, evaluation, and manuscript development.

Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach

Yongxue Xu, Ziqian Liu, Yongqing Huang, and Shuai Wang[†]

Under review at IEEE Transactions on Network Science and Engineering

Developed **DRCIM-ML** for robust seed selection in competitive multilayer networks under node-removal attacks, using diffusion-aware role alignment and cross-layer evolutionary search.

Extended RCIM from single-layer to multilayer networks, with a new formulation and a dedicated solver.

Contribution: Led problem formulation, method design, experiments, and writing; one patent pending.

Identifying Critical Nodes with Deep Learning and Reinforcement Learning: A Case Study on Urban Road Networks

Zhonghan Liu, Wenshuo Liu, Yongxue Xu, Ziqian Liu, Yichen Liu, and Shuai Wang[†]

GBCESC 2025 / [Best Paper Award](#)

Studied critical-node identification in urban road networks through comparative evaluation of graph neural network and deep reinforcement learning approaches.

Contribution: Dataset construction, GNN/DQN baseline reproduction, and experimental visualization.

^{*} Equal contribution. [†] Corresponding author.



Research Tools

AstraDraw. Developed a reference-guided workflow for editable PowerPoint research figures, reusable visual assets, and iterative local refinement. Used in my own papers; attracted [100+ GitHub stars](#).

CCFA-Skills. Co-developed with [InkMind.AI collaborators](#): a coordinated toolkit of 17 research skills spanning literature, experiments, writing, figures, review, and submission. Integrated AstraDraw for reference-guided figure authoring.



Research Initiatives & Collaborations

Collaborate with researchers at [HKUST](#), [SJTU](#), and [Westlake University](#) on video generation and multimodal spatiotemporal understanding; explore [AI-assisted research](#) with peers in [InkMind.AI](#).



Awards & Competitions

Sun Yat-sen University First-Class Scholarship
2023–2026

Chinese Mathematics Competition (CMC), Guangdong Region – Second Prize
Dec. 2024

Mathematical Contest in Modeling (MCM), USA – Honorable Mention
Feb. 2025

National Undergraduate Smart Car Competition, South China Region – Second Prize
Aug. 2025

Beyond Research: Table tennis, music, and nature travel.